



SHIVANSH GARG



ACADEMIC DETAILS

Year	Degree / Board	Institute	GPA / Marks(%)
---	M.Tech in Computer Science & Engineering	Indian Institute of Technology Delhi	8.63
2024	B.E in Computer Science and Engineering	UIET, Panjab University	9.37
2020	CBSE	Shaheed Bhai Dila Ji Public School	94.2%
2018	CBSE	Sacred Heart Convent School	97.8%

QUALIFYING EXAMS

- **Graduate Aptitude Test in Engineering (GATE) Rank: 75**

SCHOLASTIC ACHIEVEMENTS

- **Department Rank 3:** Current rank in M.Tech (Computer Science & Engineering), IIT Delhi.

IIT DELHI THESIS

Title: Atrial Fibrillation Detection Using PPG Signals in Wearable Devices

Supervisor: Prof. Rahul Garg

Description: Developing **machine learning** algorithms to enhance **Atrial Fibrillation detection** using **PPG signals from wearable devices**. This research focuses on **minimizing false positives** by distinguishing cardiac events from motion artifacts and investigating PPG-derived features to identify **novel AF patterns**, thereby improving remote cardiac monitoring precision.

PROJECTS

- **P-Wave Detection in Electrocardiograms (Prof. Rahul Garg) *Minor Project* :**
 - Performed a **systematic literature review** of **30** studies on automated P-wave detection covering signal processing, **machine learning**, and **deep learning approaches**.
 - **Implemented and validated algorithms** and benchmarked the **NeuroKit2** library, assessing overall **reproducibility**.
 - Conducted **comparative evaluation** of detection methods on **ECG databases**, analyzing performance and limitations.
- **Visual Question Answering with Multi-Modal Transformers (Prof. Parag Singla):**
 - Built a multi-modal transformer with a **ResNet image encoder** and **transformer text encoder**.
 - Used **cross-attention** fusion to improve **visual-text reasoning** on the CLEVR dataset.
 - Improved performance using **BERT embeddings and focal loss**, and validated generalization through **zero-shot**.
- **Neural Cleanse: Backdoor Attack Detection in Deep Neural Networks (Prof. Huzur Saran):**
 - Implemented and evaluated **blind backdoor attacks** on **deep neural networks** to study stealthy adversarial behavior.
 - Implemented corresponding **defense mechanisms** using **differential privacy** and **certified robustness**, and analyzed their effectiveness in mitigating attacks.
- **Text and Image Classification using Naïve Bayes and SVM (Prof. Parag Singla):**
 - Implemented **Naïve Bayes from scratch** with preprocessing and bigram based text features for text classification.
 - Built and **evaluated SVM** with linear and Gaussian kernels using CVXOPT and scikit-learn for image classification.
- **Representation Learning with Neural Networks (Prof. Rohan Paul):**
 - Implemented a custom **CNN** for 10-class bird classification with **Grad-CAM**-based feature interpretability.
 - Developed a **Variational Autoencoder (VAE)** for unsupervised representation learning and a from-scratch **Gaussian Mixture Model (GMM)** for few-shot classification.
- **Search-Based Agent for ASR Error Correction (Prof. Rohan Paul):**
 - Developed a **local search agent** to correct phonetic substitutions and missing words in speech recognition transcripts.

TECHNICAL SKILLS

- **Software and Tools :** C/C++, Python, SQL, PyTorch, scikit-learn, NeuroKit2, NumPy, Pandas, Matplotlib, xv6.

INTERNSHIPS

- **Dataware-A PKWARE Company, Mohali, Punjab (January, 2024 - June, 2024) :**
 - Worked on a project in **data security and privacy** solutions using Java.
 - Trained in **detecting and mitigating phishing** and other cyber attacks.



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IIT COURSE

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M.Tech in Computer Science & Engineering	Indian Institute of Technology Delhi	8.63	---

COURSES DONE

Advanced Data Structures, Artificial Intelligence, Intro. To Logic & Funct. Prog., Software Systems Laboratory, Teaching Practicum In Computer Science And Engineering, Machine Learning, Resource Management In Computer Systems, Networks & System Security, Minor Project, Teaching Practicum In Computer Science And Engineering